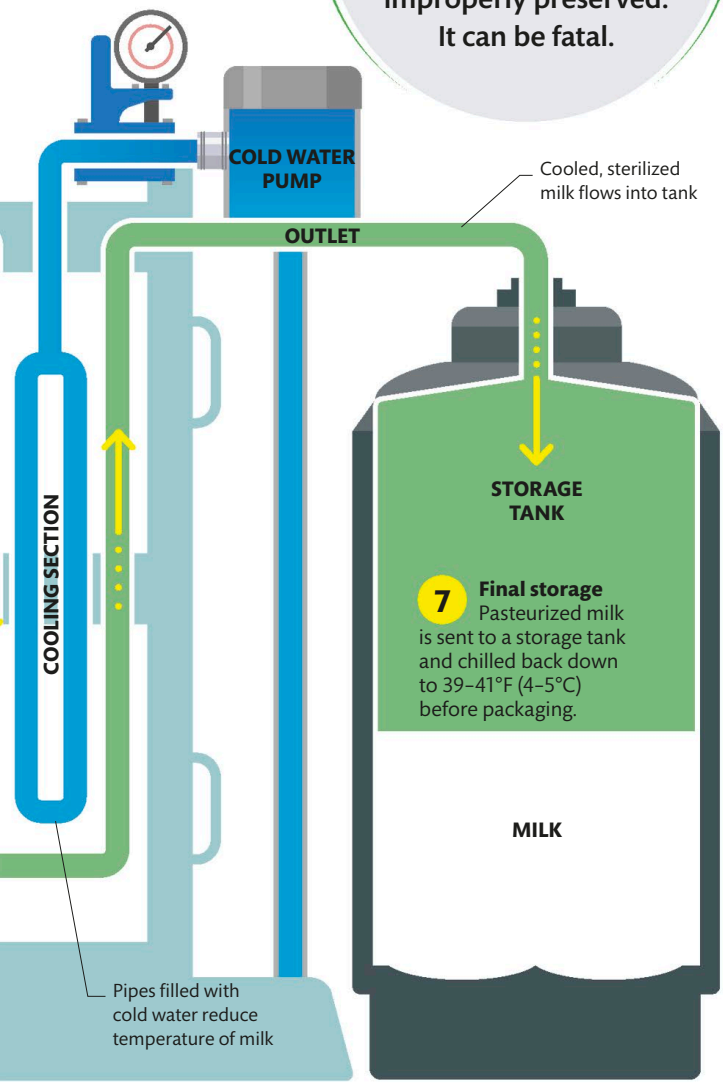
6 Flash cooling
The treated milk is then rapidly cooled in the cooling section by pipes filled with cold water provided by a cold water pump.

WHAT IS BOTULISM?

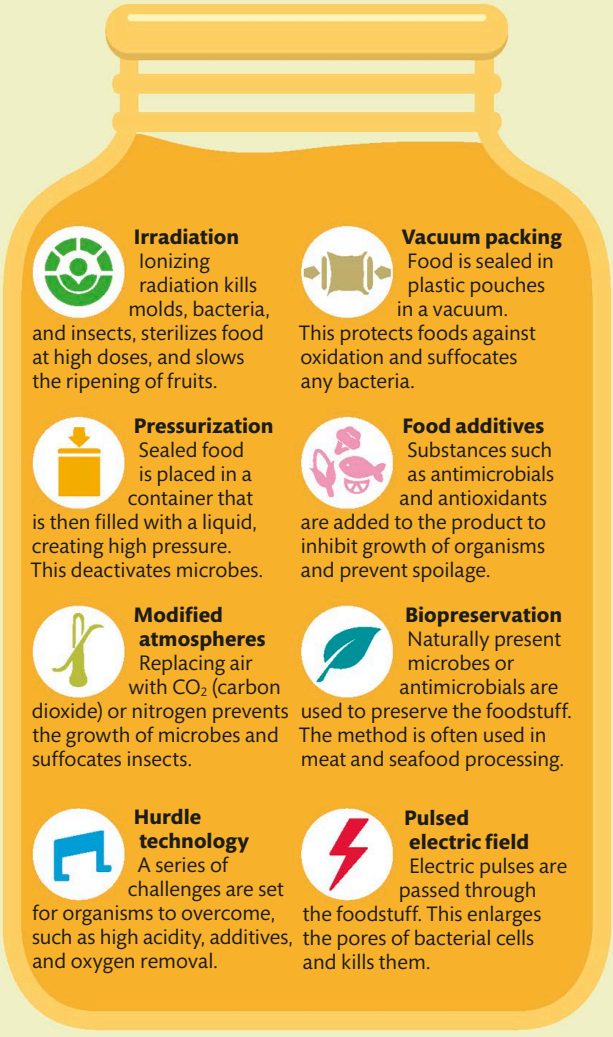
Botulism is caused by a toxin released by bacterial spores found in food that has been improperly preserved. It can be fatal.



5 Initial cooling
The heated milk travels from the holding tube to the next section of the regenerator. It is cooled by the incoming cold milk in the pipe below.

Preservation methods

Some methods of preserving food have been used since ancient times and are still in use today. Pickling, sugaring, fermentation, smoking, curing, cold storage, salting, freezing, canning, and even burial all create conditions that are hostile to spoilage organisms. In recent years, though, commercial processing has led to the development of new preservation technology.



Irradiation
Ionizing radiation kills molds, bacteria, and insects, sterilizes food at high doses, and slows the ripening of fruits.

Vacuum packing
Food is sealed in plastic pouches in a vacuum. This protects foods against oxidation and suffocates any bacteria.

Pressurization
Sealed food is placed in a container that is then filled with a liquid, creating high pressure. This deactivates microbes.

Food additives
Substances such as antimicrobials and antioxidants are added to the product to inhibit growth of organisms and prevent spoilage.

Modified atmospheres
Replacing air with CO_2 (carbon dioxide) or nitrogen prevents the growth of microbes and suffocates insects.

Biopreservation
Naturally present microbes or antimicrobials are used to preserve the foodstuff. The method is often used in meat and seafood processing.

Hurdle technology
A series of challenges are set for organisms to overcome, such as high acidity, additives, and oxygen removal.

Pulsed electric field
Electric pulses are passed through the foodstuff. This enlarges the pores of bacterial cells and kills them.

GRAIN STORED IN A **CO_2 ATMOSPHERE** REMAINS EDIBLE FOR **FIVE YEARS**